

User manual – WiRoc

Wireless split times for orienteering

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1. Overview

WiRoc is a device for wirelessly sending split times from controls to the competition software using LoRa technology. A WiRoc device can be configured as a LoRa transmitter, LoRa receiver or LoRa repeater. The device is battery-powered and rechargeable via USB.



Ethernet/USB side



Serial port side

Specifications

Dimensions	Approx. 83 × 58 × 35 mm (excluding antenna and cables)
Weight	Approx. 260 g
Battery	Built-in rechargeable LiPo battery. Operating time approx. 15 hours (without external USB devices). Charged via the USB cable (USB A).
LoRa radio	~440 MHz (70 cm amateur band / ISM band). Up to 12 channels on hardware version 8 (6 channels on earlier versions).
WiFi	2.4 GHz (V7 and earlier) or 2.4/5 GHz dual-band (V8).
Bluetooth	BLE 5.4 (V8), BLE 4.2 (V7 and earlier) for configuration with the WiRoc Config app.
SRR receiver	Built-in (from V7). Two channels – red and blue. Can be configured for transmission (from software v1.46).
USB ports	One USB A male connector for charging, two USB A host ports for peripherals.
Display	OLED display, 128×64 pixels.

2. Connections and parts

2.1 USB A male connector – Charging

The USB A port is used **only for charging** the batteries. The batteries can be charged while the device is running – for example with a power bank during an event if needed. Note, however, that some power banks emit a lot of radio interference, so only use high-quality power banks or charge only when needed. The built-in batteries last about 15 hours when nothing is connected to the USB ports.

2.2 USB A – Host ports

The USB A ports can be used with many different devices:

- **SportIdent unit with USB cable** – connected directly to WiRoc.
- **SportIdent SRR dongle** – to receive SRR punches (for versions with a built-in SRR receiver it is recommended to use those instead).
- **USB serial adapter** – to connect a SportIdent unit with a serial cable.
- **USB Ethernet adapter** – for a wired network connection.
- **Atheros AR9271 USB WiFi adapter** – to create a WiFi mesh between WiRoc devices.

2.3 Serial port

The serial port can be configured as both input and output. It can receive or send punches. Used to connect to:

- SportIdent units with a serial cable.
- Another WiRoc device with a cross-over cable (the two devices can then act as a relay – receive a LoRa signal on one channel and pass it on to the other device which forwards it on another LoRa channel).
- "Samplingsbox" (sunjo.uno).
- Computer with a USB serial adapter.

2.4 Antenna connector (SMA) and power button

The SMA connector is used for the LoRa antenna; the power button is located just below it.

2.5 Power button

The power button is located below the antenna connector:

- **Press 2 seconds** – start the device.
- **Press 2 seconds** (when the device is running) – start shutdown. The display shows "Shutting down".
- **Short press** (when the device is running) – browse the screens on the display.
- **Press ~10 seconds** – power to the device is cut (forced shutdown).

2.6 OLED display

Shows basic information about the device's configuration and status. Short presses on the power button let you browse between different screens showing IP address, LoRa settings, battery level and more.

2.7 SRR receiver

Built-in receiver for receiving punches via the SRR protocol. Can receive from both SportIdent SRR units and SIAC cards (touch-free). There are two receivers – one for the red channel and one for the blue channel. From software version v1.46 the SRR receiver can also be set to transmit mode to send punches another WiRoc device, or to a SRR Dongle.

2.8 LoRa LEDs

Two LEDs for LoRa are located at the edge of the circuit board. The **red LED** lights up when a message is being sent. The **blue LED** flashes when a message has been received. In normal operation you see the red one light up when a punch is sent, and shortly after the red one stops the blue one flashes when the acknowledgement (ACK) from the receiver is received.

2.9 Status LEDs

Hardware version up to and including V7: the **red LED** indicates that the WiRoc device has power. The **yellow LED** is used after boot-up to indicate status:

Hardware version V8: the **blue LED** indicates that the WiRoc device has power. The **red LED** is used after boot-up to indicate status:

- **Blink 0.25 Hz** – All WiRoc processes are running normally, battery voltage above 3.2 V and power management circuit temperature below 85 °C.
- **Blink 1 Hz** – One of the following has happened:
 - The WiRocPython systemd service is not running
 - The WiRocPythonWS systemd service is not running
 - The WiRocBLEAPI systemd service is not running
 - Battery voltage below 3.2 V (the device shuts down at 3.0 V)
 - Power management circuit temperature above 85 °C
- **No blink** – The WiRocWatchDog systemd service is probably not running.
- **Alternating between 1 Hz and 0.25 Hz** – A systemd service is probably crashing and restarting in a loop.

Normally the power management circuit only gets really hot when charging. At 85 °C the charging speed is reduced. At 90 °C the WiRoc device shuts down.

2.10 Battery master switch

A jumper on the circuit board (on the underside for V7 or earlier) acts as a master switch for the battery. Used to completely disconnect the battery for long-term storage or transport.

2.11 WiFi/BT antenna

Built-in antenna for WiFi and Bluetooth. On V8 there is support for both 2.4 GHz and 5 GHz WiFi.

2.12 Debug UART

This UART port can be used for troubleshooting. It shows Linux boot messages and provides access to the Linux terminal.

3. Starting and shutting down

Starting the device

1. Check that the battery master switch (jumper on the circuit board) is closed.
2. Connect the LoRa antenna to the SMA connector.
3. Hold the power button for about 2 seconds until the blue LED lights up (or the red one for V7 or earlier).
4. After about 30–60 seconds the device is ready. The OLED display shows the device status.
5. The status LED blinks slowly if everything is in order.

Shutting down the device

1. Hold the power button for about 2 seconds.
2. The display should show "Shutting down".
3. Wait until the display and the LEDs have gone out (takes about 10–15 seconds).

If the device hangs: hold the power button for about 10 seconds to force shutdown via the PMU. There is a very small risk of damaging the file system when forcing shutdown, so only do it when needed.

Browsing the screens

With short presses (under 1 second) on the power button you can switch between different information screens on the OLED display. The screens show, among other things:

- The device name and IP address
- LoRa channel, data rate and function (transmitter/receiver/repeater)
- Battery level
- Error messages

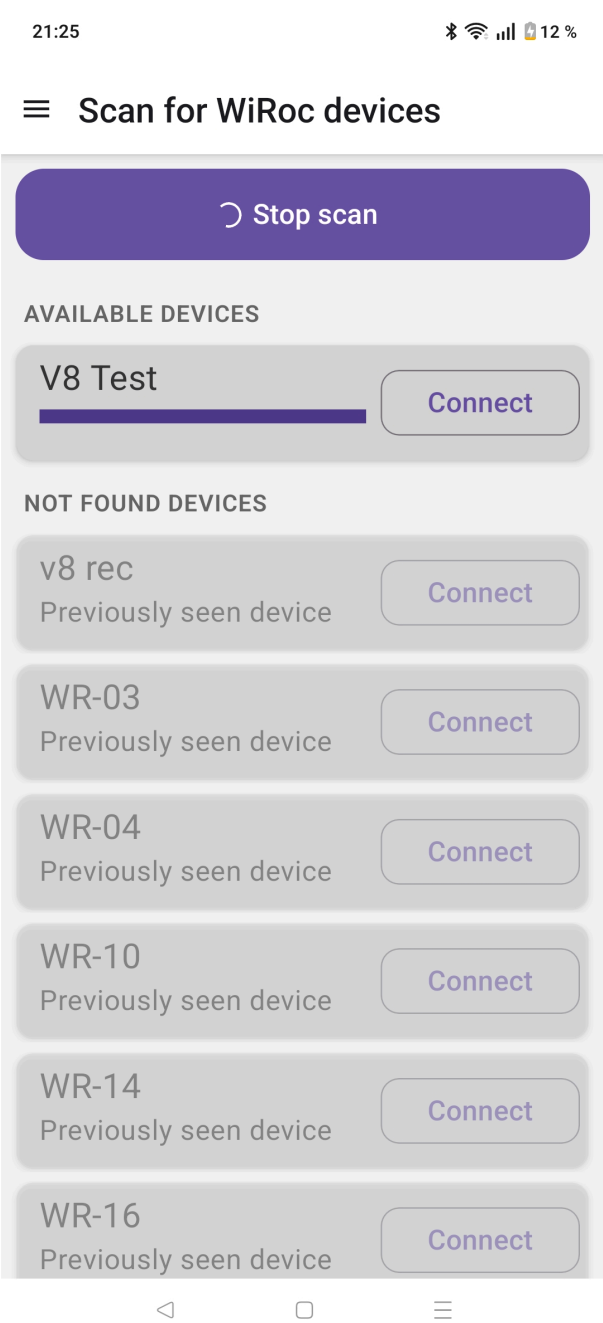
Charging the battery

Connect the USB cable to a USB port. The device can be charged both when off and when on. In normal use the battery is more than enough for a full day of competition. With a power bank the operating time can be extended if needed.

4. Configuration with WiRoc Config

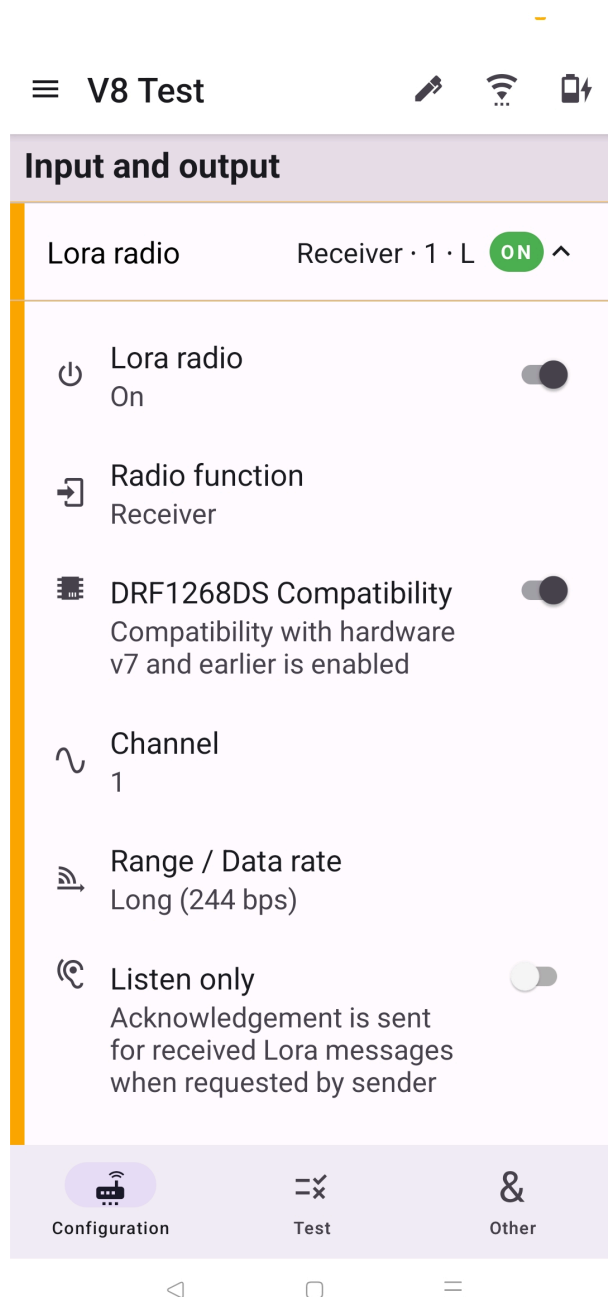
The WiRoc devices are configured with the Android app **WiRoc Config**. The app connects to the device via Bluetooth (BLE).

4.1 Scan for WiRoc devices



Scan: Start the WiRoc device and tap **Scan** in the app. Once the device has started up it appears in the list. The default name is "WiRoc Device" but it is recommended to change the name to tell them apart. If there are several with the same name, you can tell them apart by signal strength – the one with the highest signal strength is the closest.

4.2 LoRa settings



LoRa: Select function (transmitter/receiver/repeater), channel, data rate and output power. **The same channel, DRF1268DS Compatibility and Range/Data rate must be used on both transmitter and receiver for communication to work.**

The lowest data rate theoretically gives the longest range but can only handle a few punches per minute. The two lowest data rates are recommended only in specific cases, since other factors have a greater impact on range and such a low data rate comes with several drawbacks.

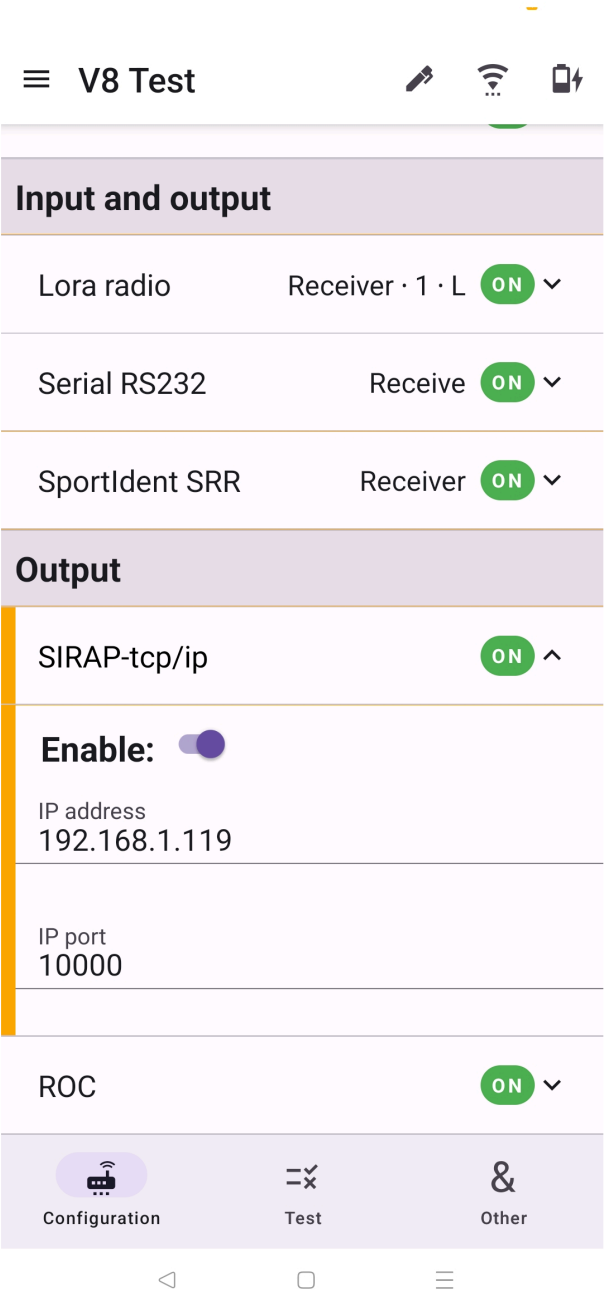
The compatibility mode for DRF1268DS (WiRoc V7 and earlier) is also set here.

With DRF1268DS compatibility mode off there are a total of 12 channels, called 1A, 1B, 2A, 2B etc. up to 6A, 6B. They are named that way to make it clear that channels 1A and 1B both overlap channel 1. That is, if you use older versions, or compatibility mode, and use channel 1, you should not use channel 1A or channel 1B at the same time.

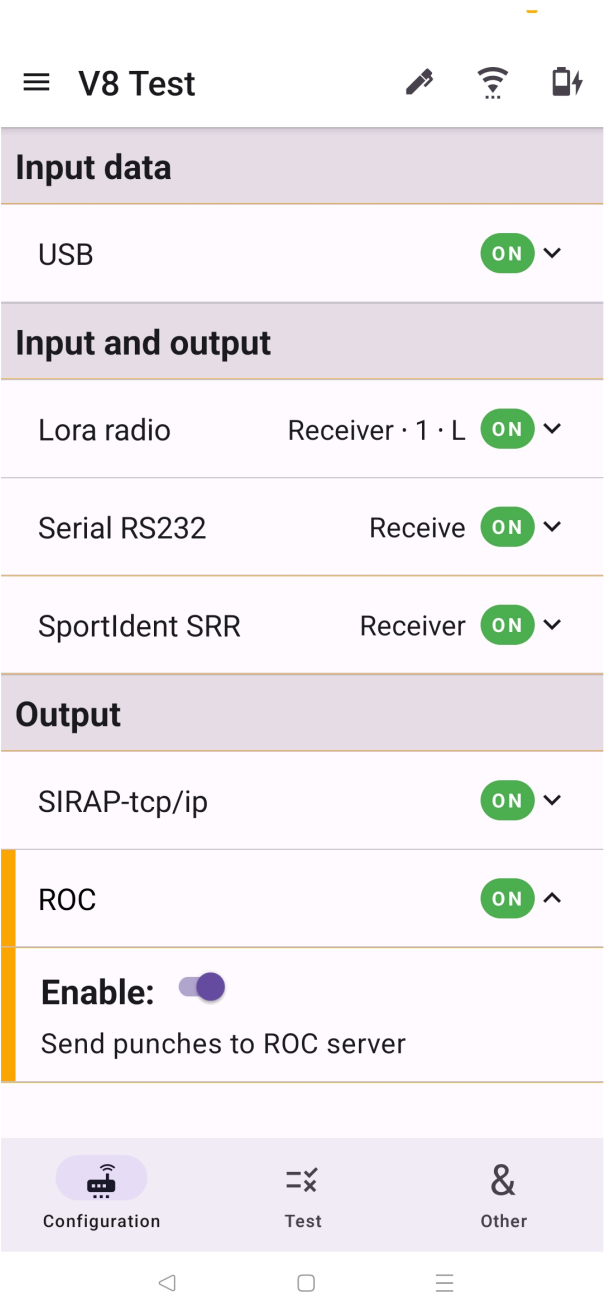
In transmit mode you can choose whether to request an acknowledgement or not; normally this should be enabled. The transmitter will then wait for an acknowledgement from the receiver and resend the message if it is not acknowledged within a certain time.

In receive mode you can choose to only listen for incoming punches/messages without sending an acknowledgement. This should normally be off (but can be used if you have another receiver that sends the acknowledgement).

4.3 SIRAP settings



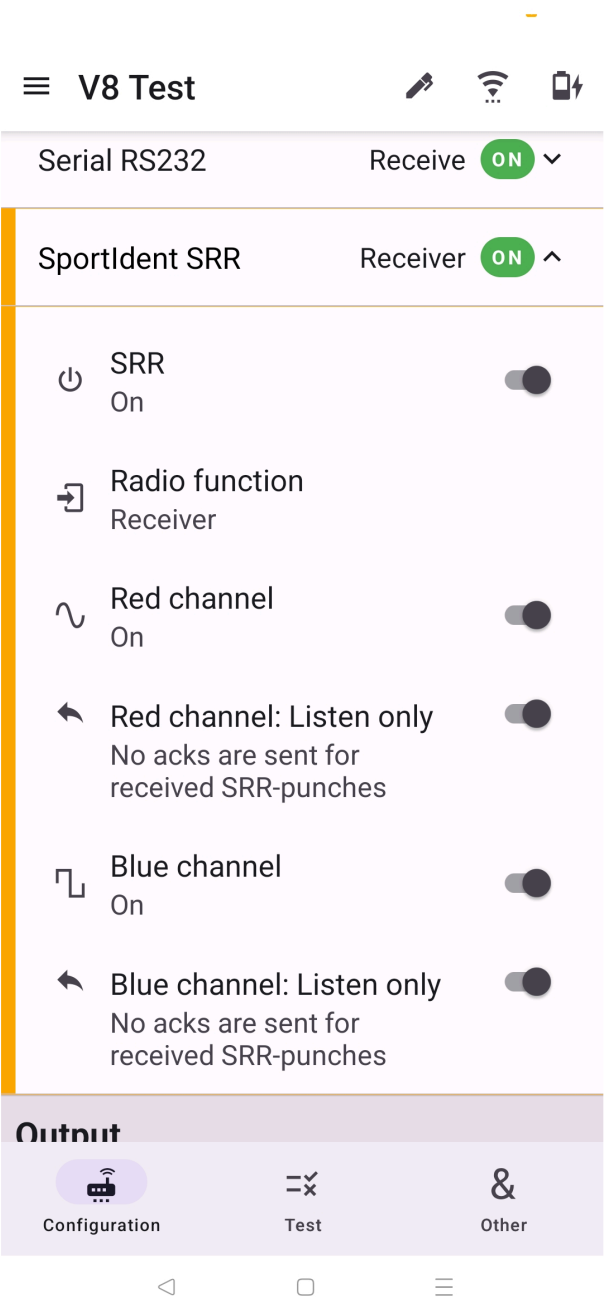
4.4 ROC settings



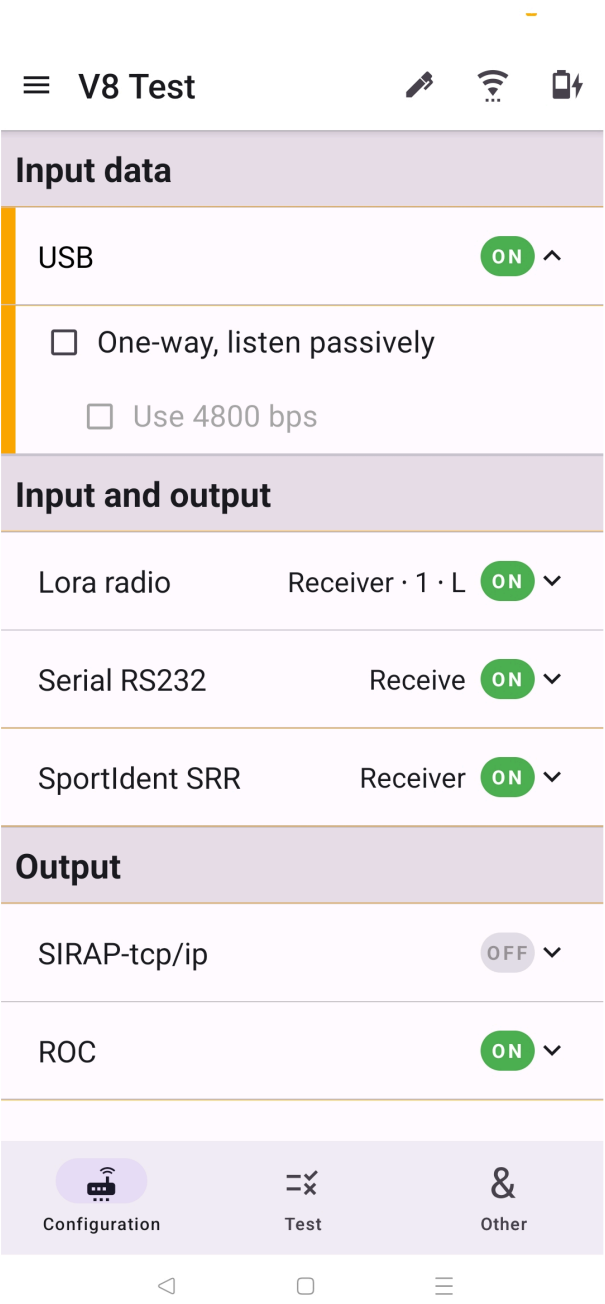
4.5 SRR settings

SIRAP: Set the IP address and port to which punches should be sent via SIRAP TCP/IP. For MeOS the default port is 10000, for OLA 10001. The receiver must be connected to the network (WiFi/Ethernet) for this to work.

ROC: Configure the device to send punches to the official ROC server. When the function is activated, incoming punches are automatically forwarded to the ROC server, just like a regular ROC device. Requires an internet connection.



4.6 USB settings

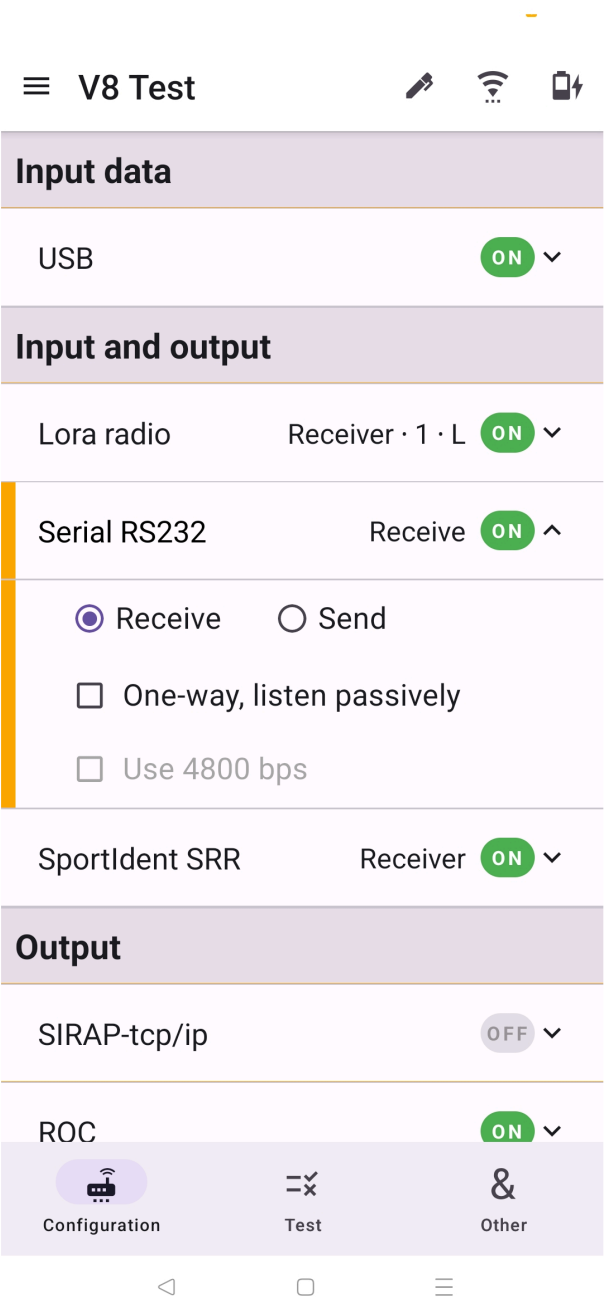


4.7 Serial port settings

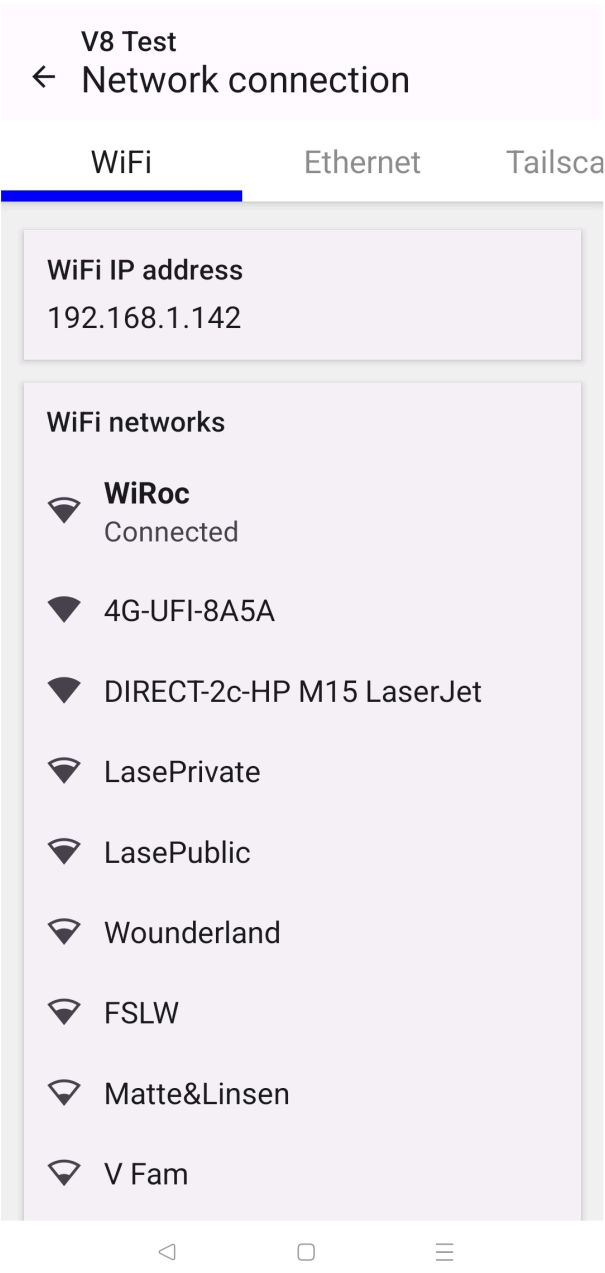
SRR: Settings for the built-in SRR receivers.

Enable/disable the red and blue channels and choose whether they should send ACK (acknowledgement) back to the SIAC/SRR units. From v1.46 the SRR can also be set to transmit mode to send out punches.

USB: Manage connected USB devices. Here you can enable "One-way" communication (listen only) for the USB ports. Can be used if you connect a "Samlingsbox" via a USB serial adapter, since the "Samlingsbox" cannot receive commands. "Samlingsbox" uses 4800 bps, so that must also be enabled.



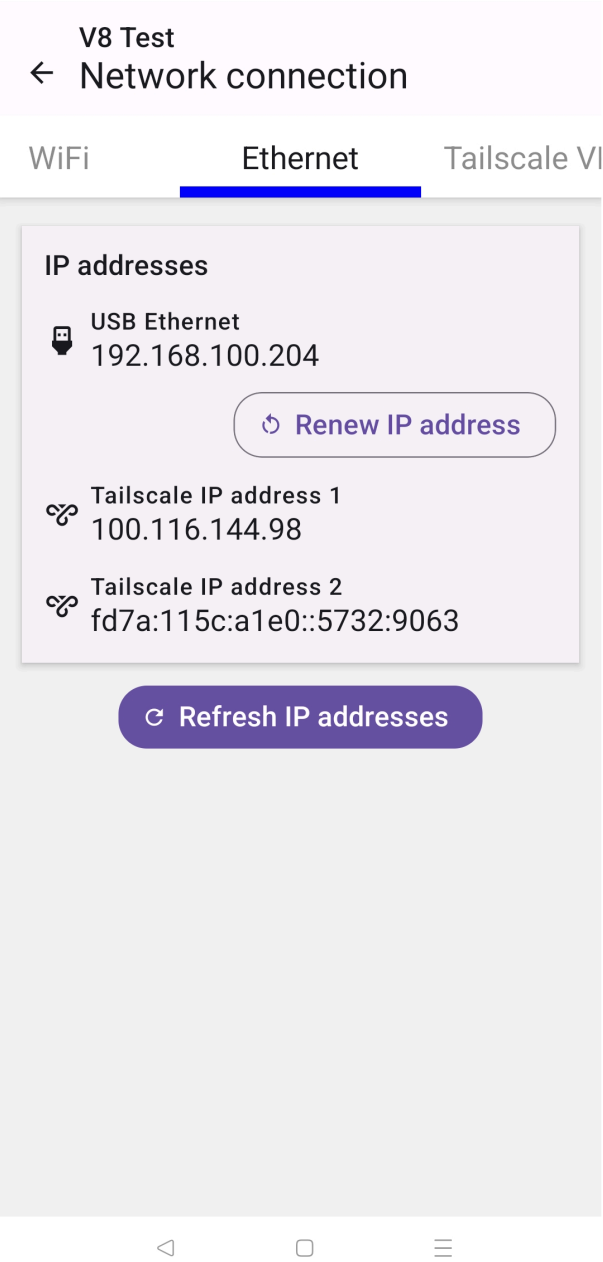
4.8 Network – WiFi



4.9 Network – Ethernet

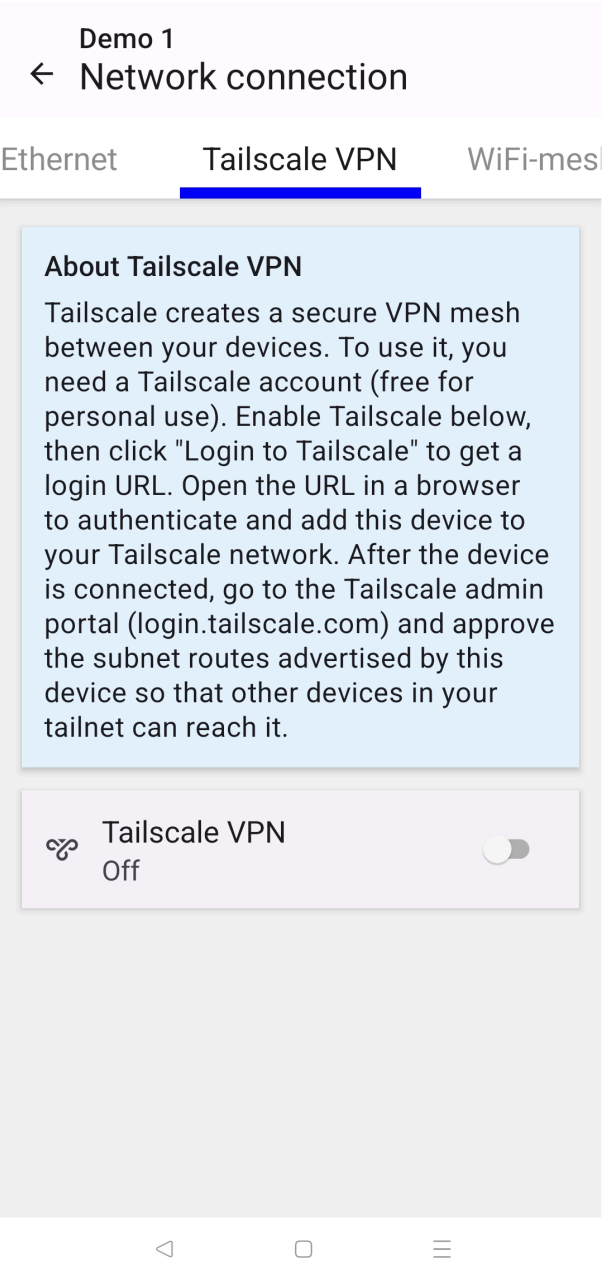
Serial port: Settings for the built-in serial port. The serial port can be used as both input (receive punches) and output (send punches). Here you can enable "One-way" communication (listen only) for the serial port when it is in receive mode. Can be used if you connect a "Samlingsbox", since the "Samlingsbox" cannot receive commands. "Samlingsbox" uses 4800 bps, so that must also be enabled.

WiFi: Connect the WiRoc device to a WiFi network. Tap **Get/Refresh wifi list** to see available networks. Select a network and enter the password. The IP address of the device is displayed when connected.



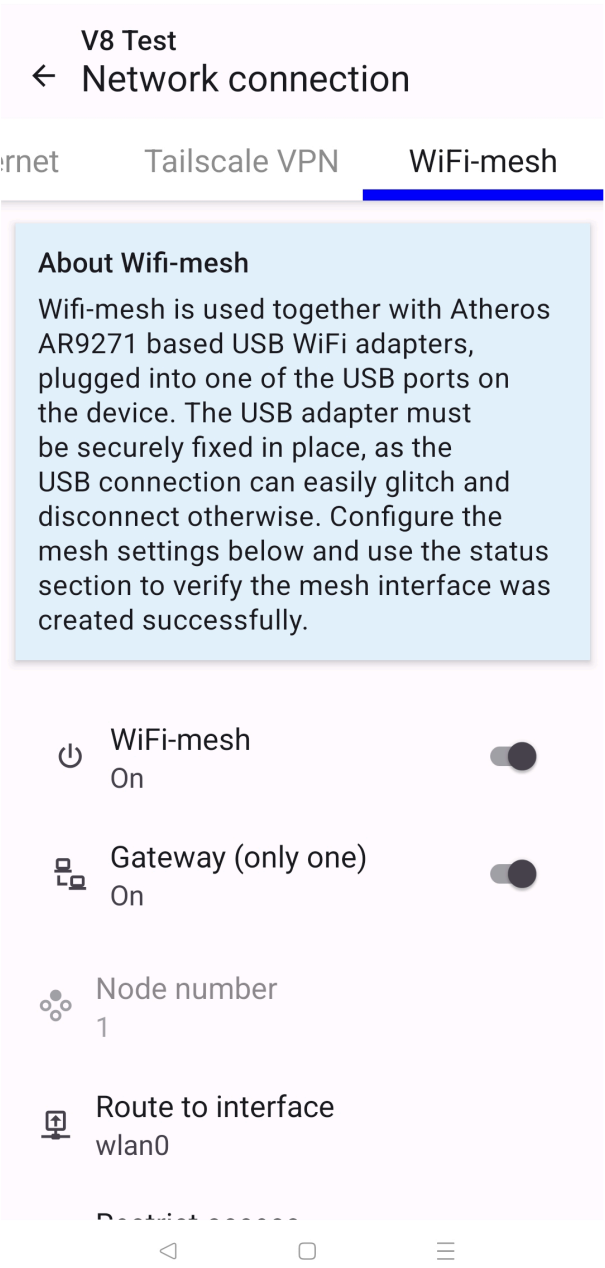
Shows the IP address for Ethernet connections (e.g. USB Ethernet adapter) and the Tailscale VPN IP.

4.10 Network – Tailscale VPN

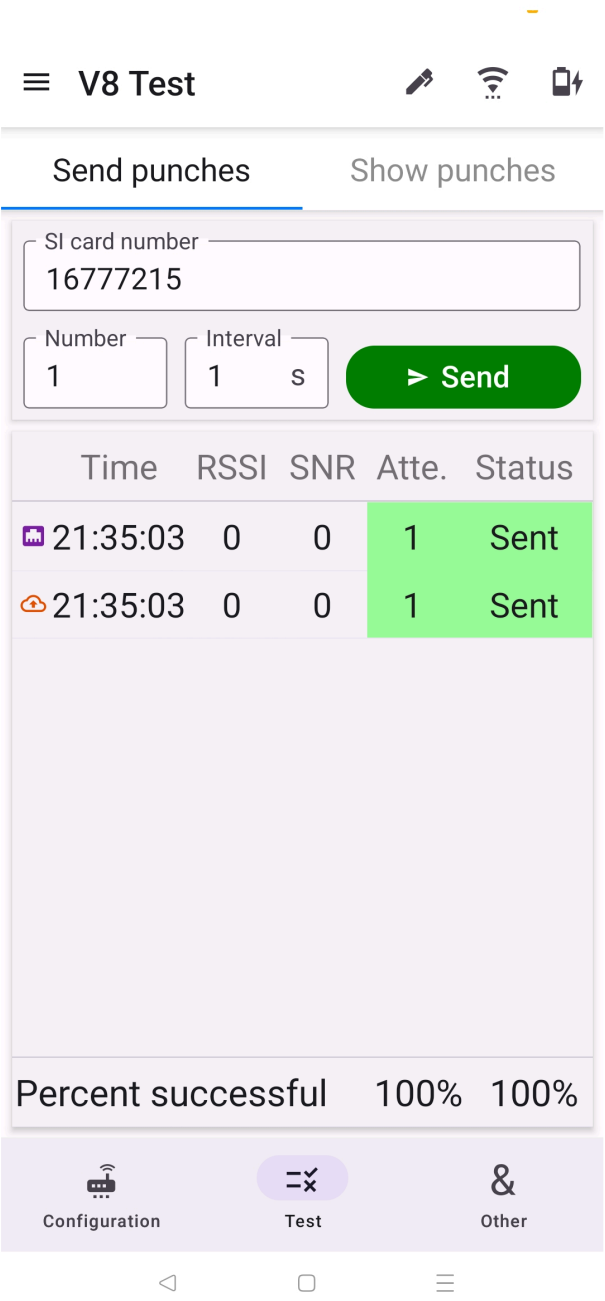


Tailscale VPN: Connect the device to a Tailscale network for remote access. The device gets a fixed IP address (100.x.x.x). Requires a free account at tailscale.com. Follow the on-screen instructions.

4.11 Network – WiFi Mesh



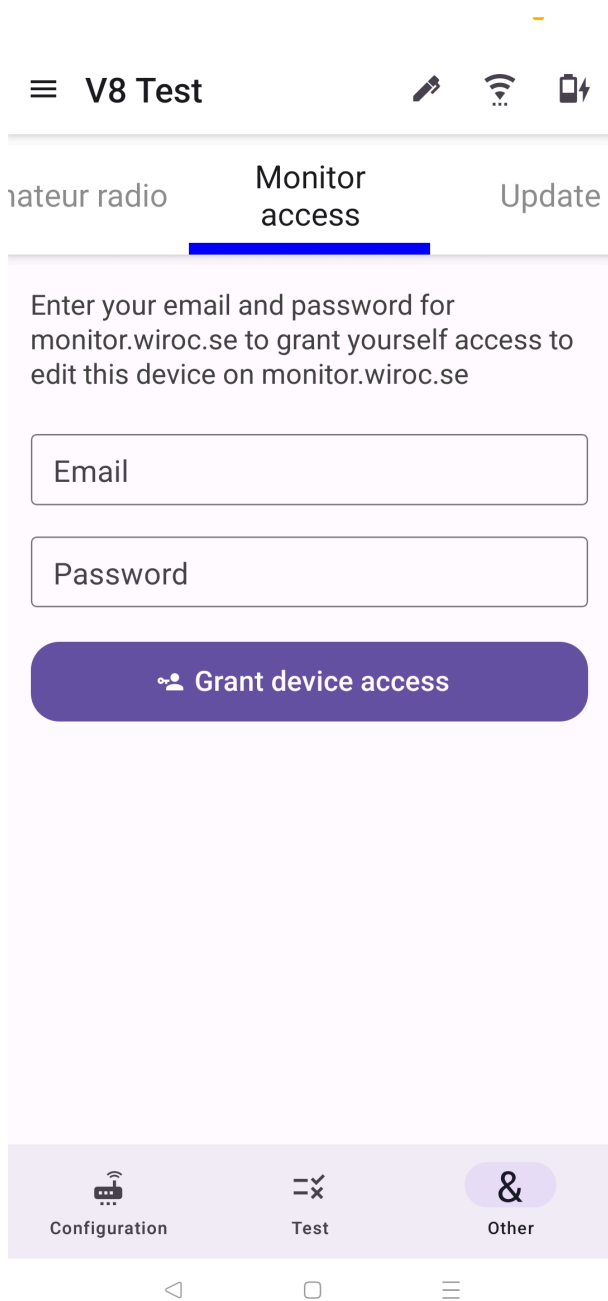
4.12 Test transmission



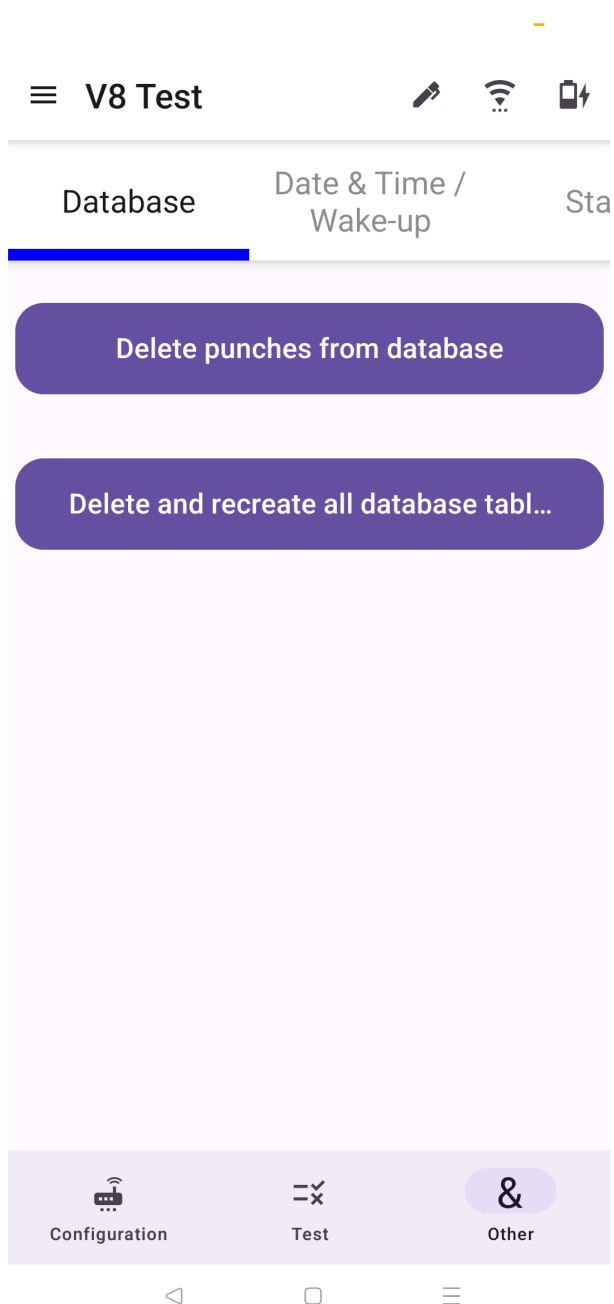
4.13 Monitor access

WiFi Mesh: Configure WiFi mesh networks (802.11s) to extend the WiFi range. Requires a compatible USB WiFi dongle (Atheros AR9271). One WiRoc device must be the Gateway and be connected to the local network. The gateway will have node number 1 and the IP address will end with the node number. On the gateway you must also select which interface the IP messages should be sent to; select the interface used to connect to the network (wlan0 if connected to WiFi). "Restrict access" can be enabled to block traffic from the mesh network to the local network. Other nodes that are not the Gateway need a unique node number.

Test transmission: Send test messages to verify the LoRa link. Shows signal strength (RSSI and SNR) and which messages were acknowledged. Used during setup to check placement and antenna direction. Can also be used to test sending to SIRAP etc.

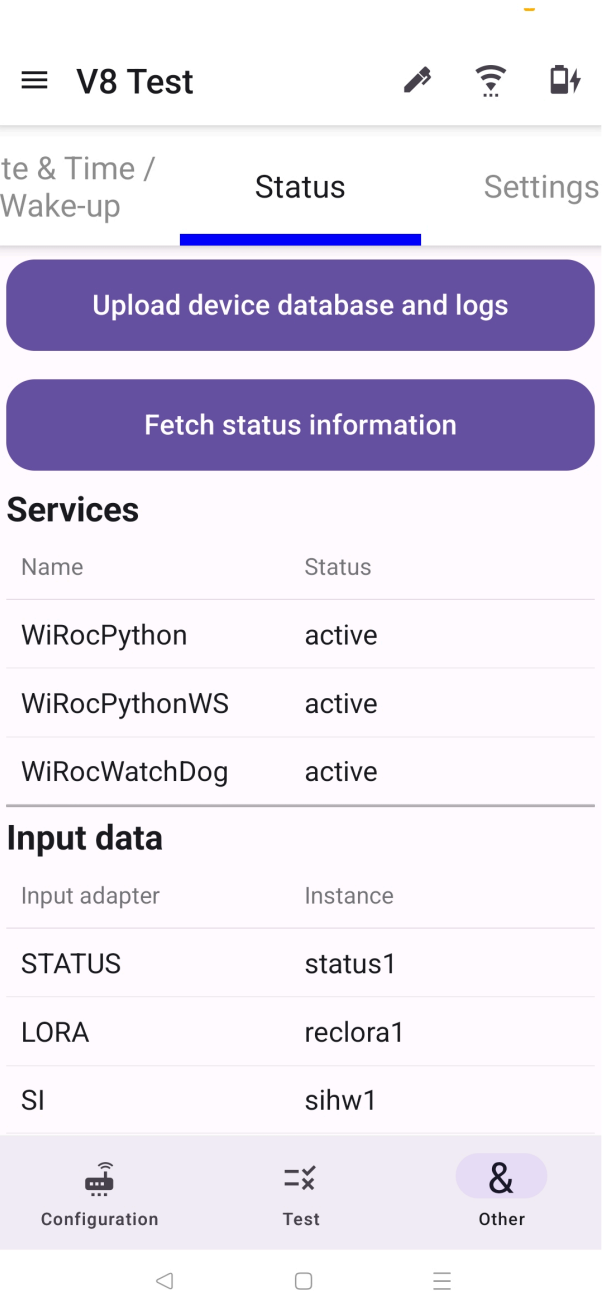
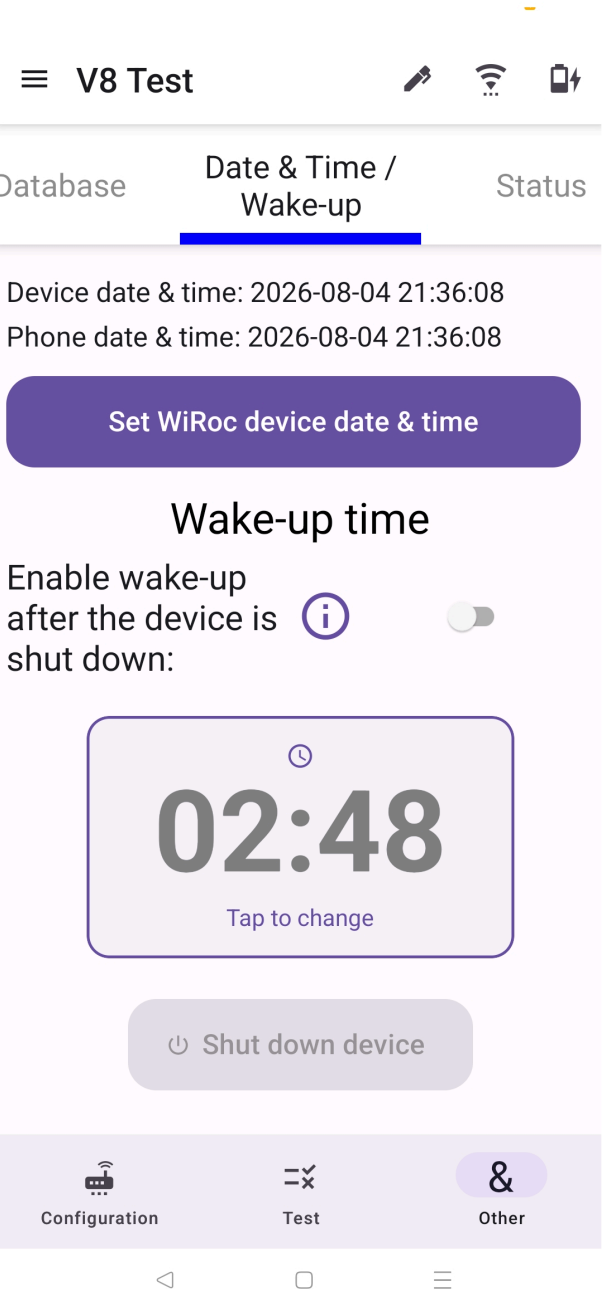


4.14 Other settings



Monitor access: Grant a username access to the WiRoc device on monitor.wiroc.se. With access, the user can assign the device to a competition, clear statistics and more. You need to create an account on monitor.wiroc.se first.

Database: Manage the device's internal database with punches and configuration. The database can be cleared of punches or completely recreated (losing both settings and punches).

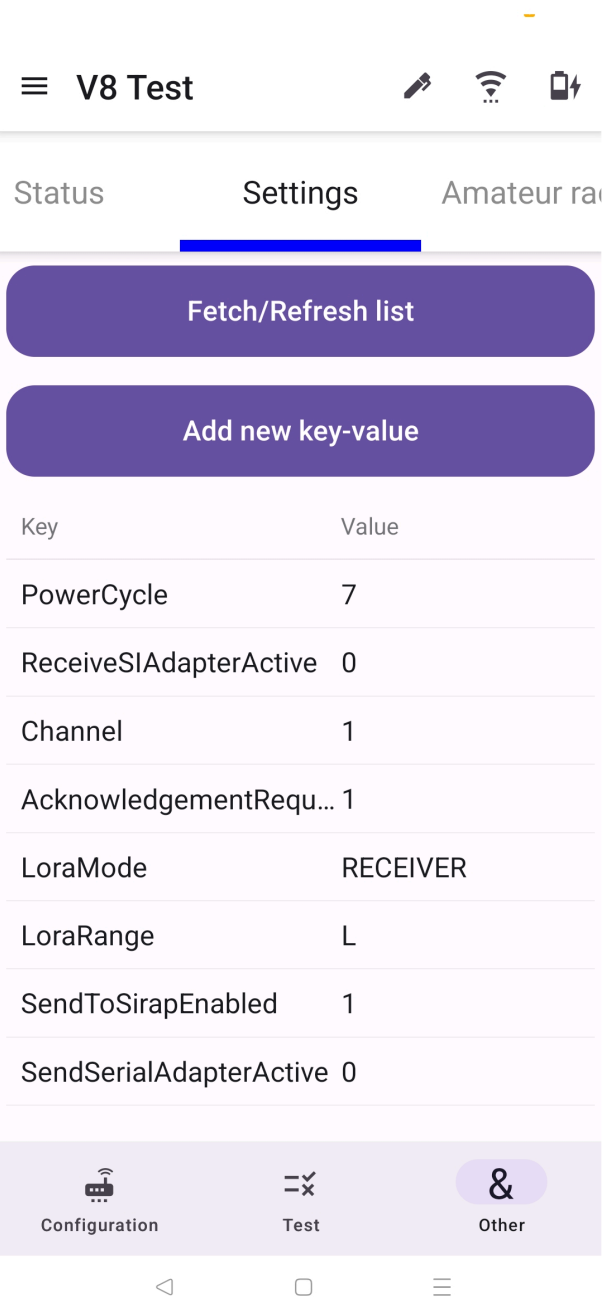


Date & Time: Set the date and time on the device.

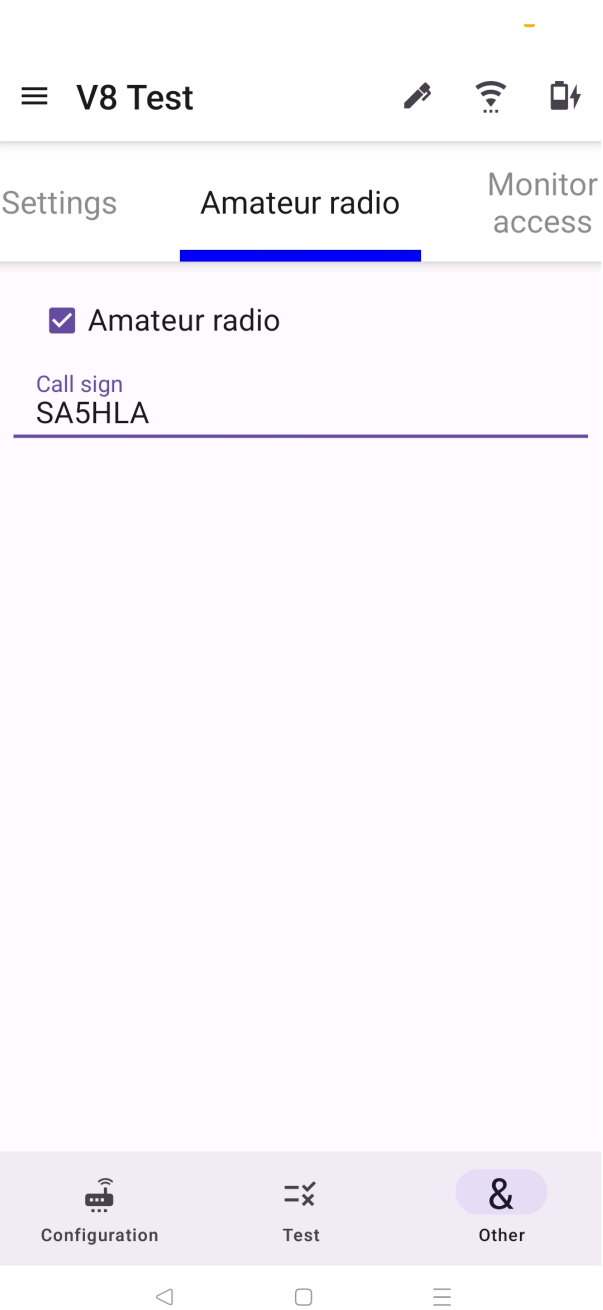
Correct time is important for logs to get the right timestamp and for punches sent to the ROC server to get the right date. The device normally synchronizes the time automatically when it has an internet connection. WiRoc V7 and later have an RTC chip that keeps the time even when the device is off.

Wake-up: Set a wake-up time for the device. After setting a time, turn off the device; it should then indicate that it has activated wake-up. At the set time the device starts up. If you start the device before the wake-up time, the wake-up is cancelled and you must configure it again if you still want it to wake up automatically.

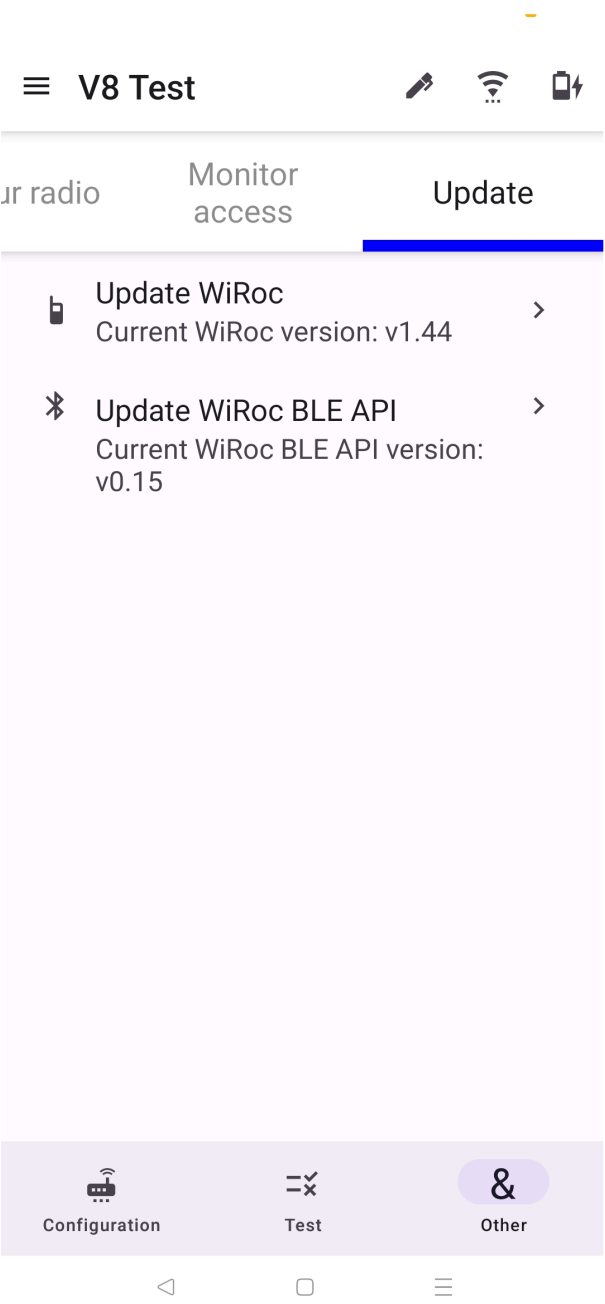
Status: Shows the current status of the device – whether the services are running as they should, which inputs/outputs are active and more. Used for troubleshooting.



Settings: Here all settings can be changed, including advanced settings that do not have a separate user interface.



Amateur radio: Settings for amateur radio mode. Here you specify whether the mode should be activated and which callsign to use for amateur radio frequencies. The callsign is sent regularly as part of the radio transmissions in accordance with applicable regulations.



Update: Update the device's software to the latest version. Keep the software updated for the latest features and bug fixes.

5. Monitor.wiroc.se

Monitor (monitor.wiroc.se) is a web-based monitoring tool for WiRoc devices. It provides an overview of all devices connected to the internet.

Getting started

1. Create an account at monitor.wiroc.se.
2. In the WiRoc Config app, under Monitor access, enter your username to give your user account the right to manage your device.
3. Once the device has an internet connection it appears in the list on Monitor.

Features

- **Device list** – Shows all WiRoc devices with battery level as a percentage, whether they are online, and last contact.
- **Detail view** – Click a device to see:
 - Graph of battery charge over time
 - Number of LoRa messages not acknowledged (orange line, 5-minute intervals)
 - Punches not acknowledged (red line, 5-minute intervals)
 - Connected USB devices and SRR dongles
 - Configuration of internal SRR receivers
- **Competitions** – Assign devices to competitions to organize and filter. Here you can also upload the map and place the WiRoc devices on the map. You can measure distances, see the elevation profile between two points, and get an estimate of how good the LoRa connection will be between the two points.

Monitoring tips

- Keep an eye on the number of unacknowledged LoRa messages – an increase can indicate deteriorating link quality.
- Check that connected SRR dongles are reported correctly. A dongle that is plugged in but not detected can cause lost punches.
- If a device goes offline – check that it hasn't turned itself off or lost contact. Online/offline status is shown based on the status messages sent every five minutes. Check whether you are receiving punches from the control in question. It could be that a single status message didn't get through.

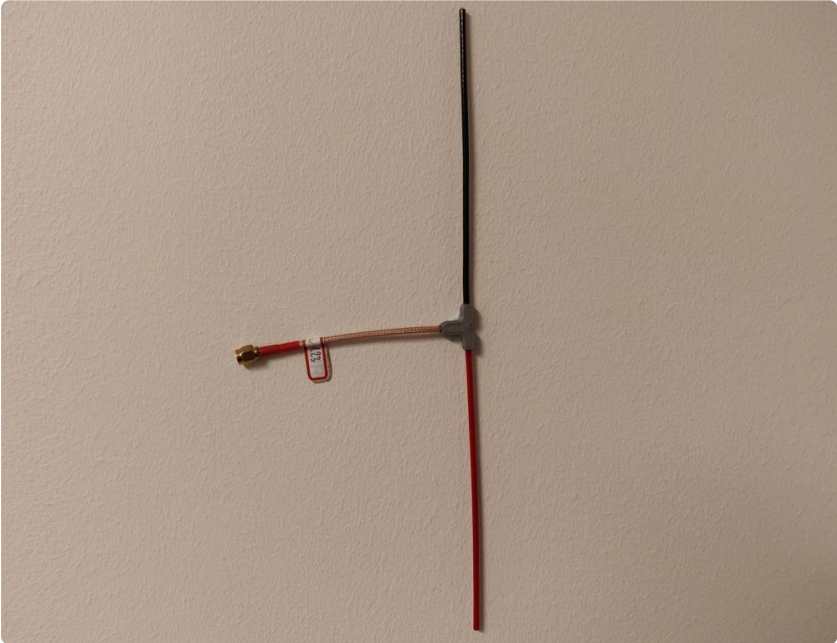
6. Range

The range of the LoRa communication depends on several factors. Here are the most important:

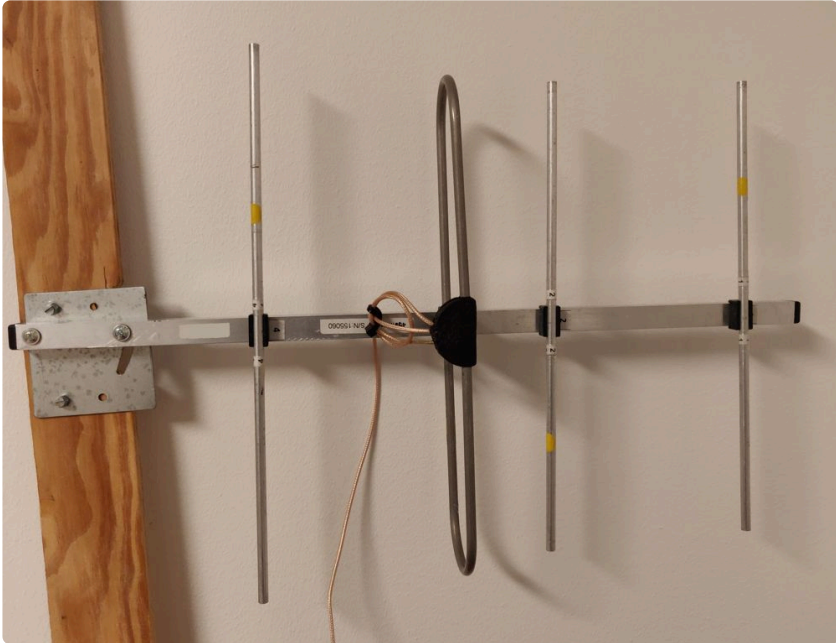
6.1 Factors affecting the range

Factor	Impact
Terrain	Free line of sight gives the best range. Hills, dense forest and buildings attenuate the signal. Place the antenna high for best results.
Antennas	A dipole antenna (included) can give good range, but a directional antenna (Yagi) can give clearly longer range. Best results are achieved with directional antennas on both transmitter and receiver. If directional antennas are used, it is important that they are pointed correctly at each other.
Antenna placement	Both the transmitter antenna and receiver antenna should have the same polarization. Usually the antenna is placed vertically. Avoid placing it near metal objects, the ground or larger objects. As clear as possible around the antenna is preferred.
Radio interference	Interference from other sources can greatly affect the range. Things that can interfere include power lines, low-quality power banks and other electronics. If power banks are used near the WiRoc device, use a high-quality one. WiRoc devices transmitting on other channels can also interfere during the moment they transmit. If two receivers are close together and one sends an acknowledgement at the same time as the other is receiving a message, it can be disrupted. Minimize the risk by placing the receivers some distance apart, using channels as far apart as possible, and reducing the risk of "collisions" by not using an unnecessarily low data rate.
Data rate	A lower data rate gives longer range but fewer punches per minute. If only one control is running on the channel, the standard (L) is a good compromise in most cases. UL and XL are recommended only in exceptional cases (very low number of punches at the control and if it doesn't work with a higher data rate).
Output power	Higher output power gives better range; the default is max and there is normally no reason to change it.

6.2 Antenna types



Dipole antenna (approx. 10×30 cm). Omni-directional. Included with rental.



Yagi directional antenna (approx. 45×30 cm). Gives significantly better range but must be pointed at the receiver.

6.3 Guidelines for good range

1. Use the test function in WiRoc Config to check whether there is contact between transmitter and receiver.
2. If possible, avoid placing the antennas so there are obstacles between them.
3. Place the antenna as high as possible – in a tree or on a pole. Place the antenna so it is as clear as possible around it; if it is placed on a tree trunk, make sure the trunk is not between the transmitter and receiver.
4. It is better to switch to a better antenna than to lower the data rate.
5. If directional antennas are used, make sure they are pointed at each other.
6. When using multiple transmitters on the same channel place them so that they can all hear each other. If they cannot hear each other then it will increase the riskt that they will transmit at the same time.
7. Remember that you have some freedom to place the receiver since it can connect over WiFi. Perhaps you can place the receiver on a nearby hill that is within WiFi range. It is also possible to run a network cable and connect the receiver via a USB Ethernet adapter (or the network port if one exists on the WiRoc device). The WiFi mesh

functionality can also be used to extend the WiFi range (but requires extra WiRoc devices and USB WiFi hardware).

8. If direct contact doesn't work – use a repeater or relay station. Place it on high ground.
9. Yet another option is to connect one of the devices, the receiver or the transmitter, to the mobile network with a 4G dongle (e.g. Huawei E3372). Using a Tailscale VPN network, connect the device to the competition's network and run SIRAP-TCP/IP to the receiver computer. For example, a receiver can be placed on high ground with good mobile coverage; transmitters located lower can then send over LoRa to the receiver and do not need mobile coverage themselves. If you don't want to use Tailscale VPN, the device can instead send the punches to the ROC server. A receiver computer must then be configured to fetch from the ROC server.